4 A cyclotron uses a magnetic field to maintain alpha particles in nearly circular paths. Electromagnets with ferrous cores provide the required strong magnetic field as shown in Fig. 4.1.

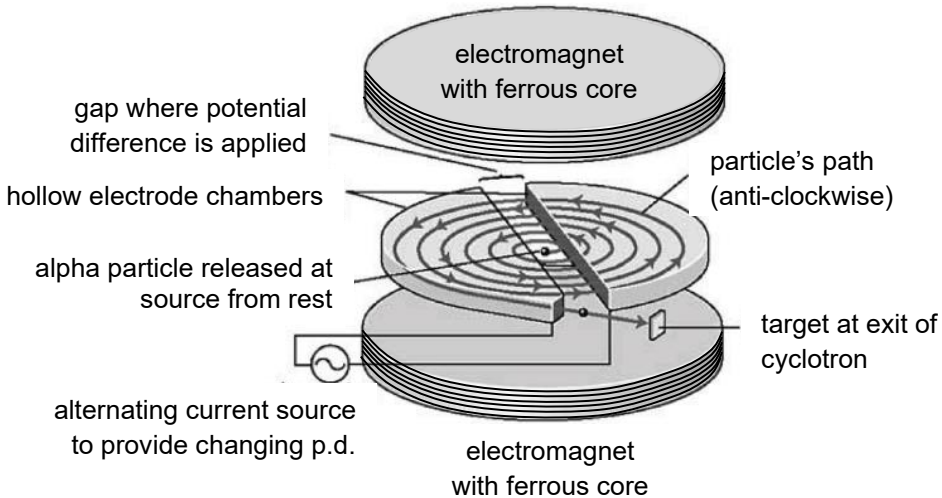

Fig. 4.1

The ions, originating from a source at the centre, released from rest, move within two D-shaped hollow electrode chambers called *dees*, separated by a very narrow gap.

Each time the ions pass into the gap between the *dees*, they are subjected to a potential difference applied between the *dees*, subsequently increasing the radius of curvature of their path. After many revolutions, the ions acquire high speeds and reach the outer edge of the cyclotron.

(a) State what is meant by a *magnetic field*.

……..……

……..……... [1]

(b) On Fig. 4.1, draw arrows to indicate the direction of magnetic flux density provided by the electromagnets.

[1]

(c) **(i)** By considering the path taken by the alpha particles and forces acting on them in the cyclotron, derive an expression for the radius of curvature *r* of the path. Express your answer in terms of mass of ions, charge, linear speed *v*, and magnetic flux density *B*. Charge should be in terms of elementary charge *e* and mass in terms of unified atomic mass constant *u*. Show your working clearly.

[2]

(ii) Hence, explain why the radius increases each time after the ions pass between the *dees*.

……

……

…….. [1]

(d) The magnetic flux density in the dees provided by the electromagnets is equivalent to that at the centre of a solenoid of length 0.39 m, 3100 turns, and a current of 200 A flowing in it.

Show that the magnetic flux density in the dees is 2.0 T.

[1]

(e) The potential difference between *dees* varies periodically as shown in Fig. 4.2.

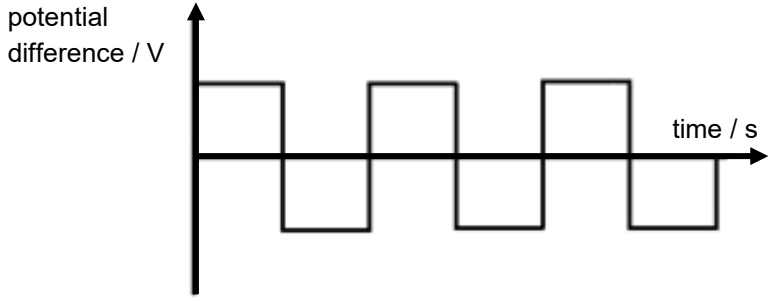

Fig. 4.2

The alpha particles reach an energy of 16 MeV after accelerating within the cyclotron for a duration of 0.52 ms and exiting the cyclotron at a radius of 0.29 m. Determine the potential difference between the *dees* at each crossing.

potential difference = …… V [4]